

§1 Inradius Boundary Inequalities in Metric Geometry

This section explores the optimization boundaries connecting a triangle's side configurations to the structural limits imposed by its inscribed circle diameter.

Problem Statement

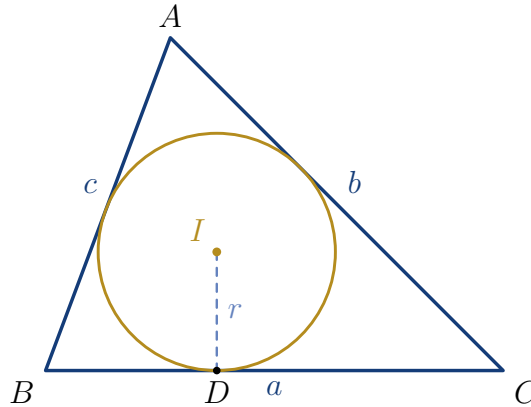
In a triangle $\triangle ABC$ with its side lengths a , b , and c , where d is the diameter of its inscribed circle, prove that:

$$d^2 + (a - b)^2 < c^2$$

Source: Classical Euclidean Inequalities

1.1 Geometric Visualization

The diagram below illustrates the geometric components under consideration: a scaling triangle with side bounds a, b, c , its incircle of diameter $d = 2r$, and the relative metric projecting along side c .



1.2 Rigorous Algebraic Proof

Proof. Let $s = \frac{a+b+c}{2}$ denote the semi-perimeter of $\triangle ABC$, and Δ represent its area. The inner metric dimensions are fundamentally tied to the inradius r via structural area properties.

By analyzing the area relationships, we can establish an exact definition for the incircle's diameter. Combining Heron's formula with the standard inradius area link gives:

$$\Delta = rs = \sqrt{s(s-a)(s-b)(s-c)}$$

Squaring both sides allows us to cleanly isolate the square of the inradius:

$$r^2 s^2 = s(s-a)(s-b)(s-c) \implies r^2 = \frac{(s-a)(s-b)(s-c)}{s}$$

Since the inscribed circle's diameter is defined as $d = 2r$, squaring it yields $d^2 = 4r^2$. This allows us to represent the diameter entirely in terms of the triangle's semi-perimeter and side dimensions:

$$d^2 = \frac{4(s-a)(s-b)(s-c)}{s}$$

To simplify the numerator, we can break down the semi-perimeter subtraction terms into their direct side-length values:

$$s-a = \frac{b+c-a}{2}, \quad s-b = \frac{a+c-b}{2}, \quad s-c = \frac{a+b-c}{2}$$

Substituting these fractional representations back into the numerator of our expression for d^2 reveals a highly symmetrical relationship:

$$d^2 = \frac{4 \cdot \left(\frac{b+c-a}{2}\right) \left(\frac{a+c-b}{2}\right) \left(\frac{a+b-c}{2}\right)}{\frac{a+b+c}{2}}$$

$$d^2 = \frac{(b+c-a)(a+c-b)(a+b-c)}{a+b+c}$$

With the diameter equation fully expanded, we can turn our attention back to our target inequality. Rearranging the terms dynamically isolating d^2 gives us a clear target boundary to test:

$$d^2 < c^2 - (a-b)^2$$

The expression on the right side matches the difference of squares identity perfectly. Factoring it yields:

$$c^2 - (a-b)^2 = [c - (a-b)][c + (a-b)] = (c-a+b)(c+a-b)$$

Rearranging the terms inside the parentheses allows us to look for common patterns across both sides of our relationship:

$$c^2 - (a-b)^2 = (b+c-a)(a+c-b)$$

Now, substituting both our structured representation of d^2 and this factored difference of squares back into the inequality produces a clean comparison:

$$\frac{(b+c-a)(a+c-b)(a+b-c)}{a+b+c} < (b+c-a)(a+c-b)$$

Based on the fundamental triangle inequality rule, the sum of any two side lengths must strictly exceed the third ($b+c > a$ and $a+c > b$). This confirms that the shared binomial factors $(b+c-a)$ and $(a+c-b)$ are unconditionally positive values.

Dividing both sides of our expression by their product safely simplifies our comparison down to a basic ratio:

$$\frac{a + b - c}{a + b + c} < 1$$

Because the geometric side lengths a , b , and c must always represent positive, real numbers, the denominator will always exceed the numerator:

$$a + b - c < a + b + c \implies -c < c$$

Since $c > 0$, this final relationship is an absolute geometric truth. Because every algebraic transition we made is fully reversible, the targeted inequality holds true for all valid, non-degenerate triangles. $\square$